

Algorithm and Dataset

Algorithm

Model Used

The Loan Approval Prediction System uses the **Logistic Regression** algorithm, a supervised machine learning classification technique. It predicts whether a loan application will be **Approved** or **Rejected** based on the applicant's details. Logistic Regression is selected because it is simple, efficient, and provides good accuracy for binary classification problems.

Step-by-Step Procedure

1. Collect the loan application dataset.
2. Load the dataset using the Pandas library.
3. Clean the dataset by removing duplicates and handling missing values.
4. Convert categorical values (such as Gender, Education, and Property Area) into numerical values using encoding techniques.
5. Normalize or scale numerical features if required.
6. Split the dataset into training data (80%) and testing data (20%).
7. Train the Logistic Regression model using the training dataset.
8. Evaluate the model using the testing dataset and calculate its prediction accuracy.
9. Save the trained machine learning model.
10. Develop a user interface using Flask, HTML, CSS, and JavaScript.
11. Accept user input through the web application.
12. Predict whether the loan application is **Approved** or **Rejected**.
13. Display the prediction result to the user.

Logic or Method Used

The system learns patterns from historical loan application records. It analyzes important factors such as applicant income, credit history, education, employment status, loan amount, and property area. Based on these features, the trained model calculates the probability of loan approval and classifies the application as either **Approved** or **Rejected**.

Process Flow of the System

1. User enters loan applicant details.
2. Input data is validated and preprocessed.
3. The trained Logistic Regression model receives the processed data.
4. The model predicts the loan approval status.
5. The result is displayed on the web application.

Dataset

Source of Data

The dataset used for this project is the **Loan Prediction Dataset**, which is publicly available on **Kaggle**. It contains historical loan application records used for training and testing the machine learning model.

Number of Records / Samples

- Total Records: **614**
- Total Features: **13** (including the target variable)

Features or Attributes (Columns)

The dataset contains the following attributes:

- Loan_ID
- Gender
- Married
- Dependents
- Education
- Self_Employed
- ApplicantIncome
- CoapplicantIncome
- LoanAmount
- Loan_Amount_Term
- Credit_History
- Property_Area
- Loan_Status (Target Variable)

Training and Testing Data Split

- **Training Data:** 80% (approximately 491 records)
- **Testing Data:** 20% (approximately 123 records)

This split allows the model to learn from most of the data while evaluating its performance on unseen records.

Data Format or Type

- **File Format:** CSV (Comma-Separated Values)
- **Data Type:** Structured Tabular Data
- **Categorical Features:** Gender, Married, Education, Self_Employed, Property_Area, etc.
- **Numerical Features:** ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History
- **Target Variable:** Loan_Status (Approved/Rejected)

Expected Outcome

After training, the Logistic Regression model predicts whether a loan application should be approved based on the applicant's information. The system provides quick, accurate, and consistent loan approval predictions, helping banks reduce manual effort and improve decision-making.